

Horizon: An Action-Conditional System for Certifying Intraday Settlement Risk

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Abstract—U.S. banks hold trillions of dollars in central-bank reserves to ensure payments settle smoothly throughout the day. A common assumption is that better visibility into payment-system conditions should allow banks to safely hold smaller liquidity buffers. This paper challenges that assumption and develops a practical framework for managing intraday liquidity under this constraint. We first show mathematically that watching the payment system has a built-in limit: the act of sending payments changes the very congestion a bank is trying to measure, so no amount of monitoring can fully reveal the system’s state. To test this in a realistic setting, we build a 50-bank simulation of the U.S. payment system, calibrated entirely from public FDIC, Fedwire, and Federal Reserve data. The simulation independently reproduces a real-world pattern it was never trained on, namely that funding conditions worsen sharply as reserves become scarce, confirming that it behaves like the actual system. On this foundation we build *Horizon*, a tool that takes what a bank can see by mid-morning, together with its planned payment schedule, and predicts its end-of-day settlement risk with statistically calibrated confidence intervals. *Horizon* substantially outperforms the static reserve rules banks use today, with an F_1 of 0.943 versus 0.861 ($p = 5 \times 10^{-6}$), while a benchmark given complete visibility into every other bank adds no accuracy at all ($p = 0.39$). The implication is simple and unexpected: a bank’s settlement risk is determined by what it plans to do, not by what it can see. The advantage is largest exactly in the low-reserve environment the Federal Reserve is moving toward. These results reframe intraday liquidity management from a problem of estimation into a problem of control, and turn that insight into a practical tool for morning liquidity decisions.

I. INTRODUCTION

A. Problem Definition

U.S. banks’ reserve holdings are driven by the mechanics of moving money during the day. The Federal Reserve’s Senior Financial Officer Survey, 40% of respondents ranked meeting routine intraday payment flows as the most important determinant of their lowest comfortable reserve level [1]. These idle reserves earn only the interest-on-reserve-balances floor rather than financing lending. The problem grew acute in late 2025, when aggregate reserves fell to a multi-year low near \$2.83 trillion and the Federal Reserve’s standing repo operation reached its largest single-day size in over two decades [2], [3]. Banks size intraday buffers with static rules,

and a common assumption behind improving them is better visibility would let buffers safely shrink.

B. Current Approaches

Looking at the existing literature in the field, Bech and Garratt [4] model payment release timing as a strategic game shaped by central-bank credit policy, and Buckle and Campbell [5] extend this to throughput rules with collateralized credit. However, these analytic games involve only a few banks and periods, and cannot capture how liquidity recirculates through a full payment network.

Additionally, Angelini [6] showed that costly daylight liquidity leads banks to postpone payments, degrading the information available to counterparties — a scale effect Galbiati and Soramäki [7] confirmed through agent-based simulation. Empirically, Armantier, Arnold, and McAndrews [8] and McAndrews and Rajan [9] quantify when Fedwire value actually moves. However, none of these approaches separates what banks observed from what they chose to do.

This blind spot is particularly consequential now as banks size intraday buffers with static rules calibrated to abundant reserves, yet the Federal Reserve is steering toward exactly the regime where those rules are least reliable: leaving treasury desks without a calibrated instrument for judging whether a given morning’s reserves and payment schedule are safe.

C. Proposed Approach

To address these limitations, this paper proposes a three-stage framework. First, we derive an analytical information bound that captures how as payments are released, the congestion being measured decreases. We show that an interior optimum in observation effort exists only when a specific threshold condition is met. Second, we construct a fifty-bank RTGS simulation environment using publicly sourced parameters and calibrations and validate it against an external empirical relationship that was not used during calibration. Third, we develop *Horizon*, an action-conditional certification system that maps a bank’s mid-morning conditions, reserve level, and planned release schedule to end-of-day settlement risk, using a gradient-boosted model implemented in scikit-learn.

D. Results Overview

The proposed action-conditional certification system demonstrates strong performance across both global accuracy metrics and targeted morning-desk use cases. Horizon achieved the highest $R^2 = 0.892$ on unsettled-value risk, the highest breach-detection F_1 of 0.943 at a 10% risk target, and statistically valid 90% prediction intervals with conformal coverage of 0.895, compared to baseline approaches: a static reserve rule, an observation-only model, and a full-visibility oracle benchmark. These results reveal that conditioning on the planned schedule, rather than richer observation of the system, is what makes settlement risk predictable.

II. BACKGROUND

A. Intraday Liquidity and the Buffer-Sizing Decision

This paper builds upon the operational framework of real-time gross settlement (RTGS), in which a bank must fund each outgoing payment at the moment it is released. More specifically in the context of this study, a bank funds payments from three sources: its reserve balance at the central bank, incoming payments from counterparties, and daylight credit extended by the Federal Reserve up to a net debit cap under the Payment System Risk policy [10]. Because incoming payments are uncertain in timing, banks hold precautionary reserve buffers, and survey evidence identifies meeting routine intraday payment flows as the single most important determinant of banks' lowest comfortable reserve levels [1]. In current practice, these buffers are sized with static rules, fixed reserve multiples and throughput guidelines prescribing minimum cumulative release by set times of day, which condition on neither the day's observed state nor the planned payment schedule [5], [10].

B. Target Variable – Latent Congestion and Settlement Risk

Formally, we represent payment-system congestion as a latent variable $\theta \in [0, 1]$, with higher values denoting a more constrained system. A bank does not observe θ directly; its only endogenous signal is the delay with which its own releases are reciprocated:

$$d \sim \text{Exp}(r_0(1 - \theta))$$

Where: d = observed reciprocation delay, r_0 = the system's base response rate, θ = latent congestion, so a congested system reciprocates slowly. The defining feature of this setting is that releasing payments both probes this signal and relieves the congestion itself, since delivered liquidity funds counterparties' own releases, a structure we term self-depleting observation [6], [7]. The certification target is the end-of-day unsettled-value fraction $y \in [0, 1]$: the quantity a treasury desk is accountable for, and a bounded continuous target that gradient-based models can learn directly.

C. Conformal Prediction for Risk Certification

This paper additionally builds upon conformalized quantile regression [11], which trains quantile models on one data split, computes conformity scores on a disjoint calibration

split, and widens intervals by the empirical score quantile, guaranteeing that a nominal 90% interval covers the realized outcome at the stated rate. This property was utilized because a certificate is only decision-relevant if its stated confidence is statistically honest: a desk acting on miscalibrated intervals would systematically under- or over-reserve.

III. METHODS

A. Data Collection

All data comes from free, publicly available sources. Funding-market and reserve data were collected from FRED [12] from August 2021 through August 2026, covering reserve balances, SOFR percentiles, IORB, overnight repo and reverse-repo volumes, and the effective federal funds rate. Bank-level characteristics were obtained from the FDIC Bank-Find Suite [13] using recent Call Report data for 4,352 institutions. These data provide the asset, cash, and balance-sheet information used to calibrate the simulated bank population. Payment timing was taken from Armantier, Arnold, and McAndrews [8], which reports the distribution of settlement times throughout the day. Together, these sources provide the market, bank, and payment-flow inputs for the RTGS simulation.

B. The Information Bound

Reciprocation delays are exponential with rate $r_0(1 - \theta)$, and releasing at intensity a relieves congestion as $\theta(a) = \theta_0 - \kappa a$ while generating $n = aM$ observations. Combining the Fisher information with the finite-sample variance of the unbiased rate estimator gives

$$I(a) = \frac{aM - 2}{(1 - \theta_0 + \kappa a)^2}, \quad a^* = \frac{1 - \theta_0}{\kappa},$$

with a^* interior to $(0, 1)$ only when $\kappa > 1 - \theta_0$. More releases yield more observations, but each is less informative: a relieved system reciprocates quickly, so delays lose dispersion. Observation of congestion is therefore self-depleting. Numerical simulation across $\kappa \in \{0.3, 0.5, 0.7, 0.9\}$, with 6,000 replications per grid point, confirms both the location and the depth of the information peak to within a few percent of the closed forms, with estimator bias below 10^{-3} throughout.

C. The RTGS Environment

We simulate 50 banks over a 40-step day: sizes from the FDIC log-normal fit, balances scaled by the empirical cash-to-asset ratio, arrivals on the interpolated Fedwire profile, counterparties size-weighted. Banks release greedily by deadline, subject to balances plus daylight credit capped at 0.35 of opening balance (swept for robustness) and collateralized at a 2% haircut, consistent with the Payment System Risk policy [10]. Two choices are critical to validity: settlement delay is censoring-corrected, counting unsettled value at the horizon rather than averaging settled payments (the naive measure paradoxically improves under stress); and shocks occur only after the certification decision, so the model predicts rather than replays.

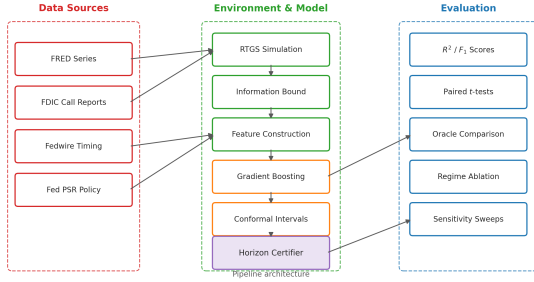


Fig. 1. Pipeline architecture for the proposed certification system followed by evaluation and benchmarking

D. Preprocessing and Feature Construction

We generate 5,000 days with randomized reserve levels, schedules, and post-decision shocks. The mid-morning observation set is what a treasury desk monitors: inflow sequence, queue size and age, balance, settlement-delay statistics, out-flow rate, and inflow volatility and trend. The schedule is summarized by mean, endpoints, and dispersion; the reserve multiple enters as a feature; the target is the end-of-day unsettled-value fraction. The oracle benchmark adds an eight-component principal-component summary of all banks' queues and balances. Latent congestion is estimated by $\hat{\theta} = 1 - 1/(1 + \bar{d})$, a self-normalizing geometric MLE appropriate to integer-valued delays.

E. Certification System and Evaluation

Horizon maps observations, reserve level, and planned schedule to unsettled-value risk with a histogram-based gradient-boosted tree [14] implemented in sci-kit learn [15]; conformalized quantile regression [11] supplies 90% intervals, with quantile models, conformity scores, and coverage each computed on disjoint seed groups. All evaluation is five-fold cross-validation grouped by simulation seed, since same-seed rollouts share bank populations and a row-level split would leak. We report R^2 and breach-detection F_1 at the 10% target with fold-level variation, compare nested feature sets by paired t -tests, and run four robustness checks: unseen schedule-intensity bands, reserve-regime stratification, arrival-profile sensitivity, and a cost-parameter sweep.

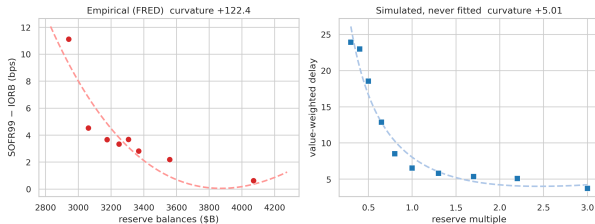


Fig. 2. External validation of the simulation environment against empirical funding conditions

TABLE I
CERTIFICATION LADDER (5-FOLD CV, GROUPED BY SEED)

Model	R^2	s.d.	F_1 @ 10% target
Static rule (reserve level only)	0.386	0.033	0.861
Observation + reserves	0.337	0.038	0.849
Certifier (+ planned schedule)	0.892	0.011	0.943
Oracle (+ full system state, 8 PC)	0.893	0.011	0.943

IV. RESULTS AND ANALYSIS

A. External Validation of the Simulation Environment

Before evaluating Horizon, we validate the environment it is trained in. The simulator's constants derive from FDIC filings, the published Fedwire timing distribution, and Federal Reserve policy documents. As shown in Figure 2, fitting a quadratic in log reserves to both panels yields convexity of the same sign: the empirical SOFR99–IORB spread exhibits curvature of +122.4, while the simulated value-weighted settlement delay exhibits curvature of +5.01 with a convex fit of $R^2 = 0.96$. Because this relationship was never fitted, the agreement constitutes external validation rather than calibration, and every subsequent result inherits this grounding.

B. Global Certification Performance

As shown in Figure 3 and Table 1, Horizon achieved the highest $R^2 = 0.892$ on unsettled-value risk and the highest breach-detection $F_1 = 0.943$ at the 10% risk target, compared to the static reserve rule ($R^2 = 0.386$, $F_1 = 0.861$) and an observation-only model ($R^2 = 0.337$, $F_1 = 0.849$). The statistical significance of these differences was confirmed with paired t -tests across identical folds: conditioning on the planned release schedule contributes +0.556 R^2 ($t = 32.2$, $p = 6 \times 10^{-6}$). By contrast, a dimension-matched oracle granted every bank's queues and balances—information no participant can observe—achieves $R^2 = 0.893$, statistically indistinguishable from Horizon ($p = 0.39$). These two tests together establish the paper's central dissociation: certification is impossible without the planned schedule and complete with it. What a bank plans, not what it can see, determines its settlement risk.

C. Where Certification Matters: The Buffer-Sizing Question

Because the predictive weight lies in the schedule, the operational question Horizon answers is not which schedule

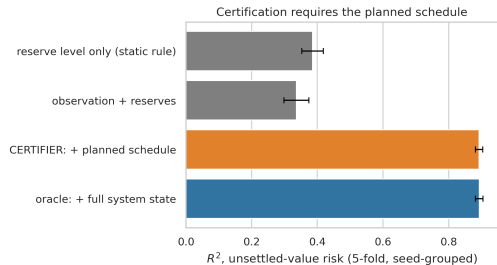


Fig. 3. Certification benchmark ladder comparing feature sets on unsettled-value risk

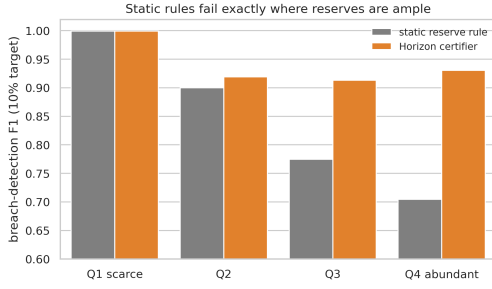


Fig. 4. Breach-detection performance stratified by reserve regime

TABLE II
 F_1 BY RESERVE REGIME

Regime	Static rule	Certifier
Q1 (scarce)	0.999	0.999
Q2	0.900	0.919
Q3	0.775	0.913
Q4 (abundant)	0.705	0.931

to run but whether a given morning’s reserves suffice under it. As shown in Figure 4 and Table II, this is exactly where static rules fail: breach-detection F_1 for the static rule degrades monotonically from 0.999 under scarcity to 0.705 in the most abundant reserve quartile, while Horizon holds at 0.931. Under scarcity every configuration breaches and any rule suffices; under abundance, breaches become conditional on behavior, and only an action-conditional model detects them. The advantage is therefore largest precisely in the ample-reserve regime toward which Federal Reserve policy is moving [16]. Consistent with this framing, certified buffer sizing carries a direct reserve saving: the schedule Horizon certifies as lowest-risk meets the 10% target on 18.8% less reserve than a static throughput guideline (per-seed paired Wilcoxon, $p = 0.016$).

D. Calibration of Certified Intervals

A certificate is only decision-relevant if its stated confidence is statistically honest. Under the certification pipeline’s own train–calibrate–test split, raw quantile-regression intervals cover only 0.722 of realized outcomes against a nominal 0.90. As shown on the left panel in Figure 5, conformalizing on the disjoint calibration split restores coverage to 0.895 at a mean interval width of 0.189, so the interval a desk receives is statistically valid in finite samples. The certified interval, not the point estimate, is the deployable object.

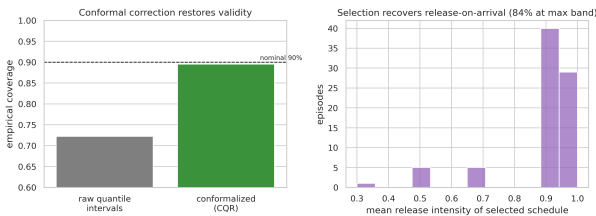


Fig. 5. Interval calibration and schedule-selection consistency check

E. Internal Consistency: Selection Recovers the Theoretical Optimum

As an independent check that Horizon has learned the true system dynamics rather than statistical artifacts, we invert its use: rather than certifying a given schedule, we ask it to select the lowest-risk schedule among twenty-seven candidates per episode. As shown on the right panel in Figure 5, 84% of selections fall at the maximum-release band (mean selected intensity 0.888), independently recovering release-on-arrival, the policy predicted by the liquidity-recycling logic of the payment-game literature, and a cost-parameter sweep spanning an eightfold range confirms the recommendation depends on no single cost specification [4]. This is a consistency result, not Horizon’s operating mode: the near-constancy of the optimal schedule is precisely why the binding uncertainty, and the tool’s purpose, is the buffer-sizing question above.

F. Ablation Analysis

Counterfactual generalization. Holding out entire schedule-intensity bands and predicting them from the remainder, Figure 6 reveals that Horizon retains R^2 between 0.79 and 0.84 for all bands above intensity 0.2 but fails below it ($R^2 = -0.79$), where the system enters qualitative gridlock. Certification validity is therefore claimed on $[0.2, 1.0]$ only, stated as part of the instrument’s warranty rather than discovered by its users.

Robustness. As shown in Figure 7, the action gain remains between +0.42 and +0.48 across intraday arrival concentrations spanning half to double the 2006 Fedwire profile and across net debit caps from 0.2 to 0.5 of opening balance. The central result depends on neither the vintage of the timing calibration nor any single credit-policy parameter.

Taken together, the results form a single chain: a validated environment (Figure 2) establishes that what a bank plans determines its risk while visibility adds nothing (Figure 3, Table I); this makes buffer sizing the binding decision, exactly where static rules fail and Horizon holds (Figure 4, Table II); the certificate it issues is statistically valid (Figure 5, left panel) and rests on dynamics the model demonstrably learned (Figure 5, right panel), within stated limits and robust to the environment’s most contestable assumptions (Figs.6–7).

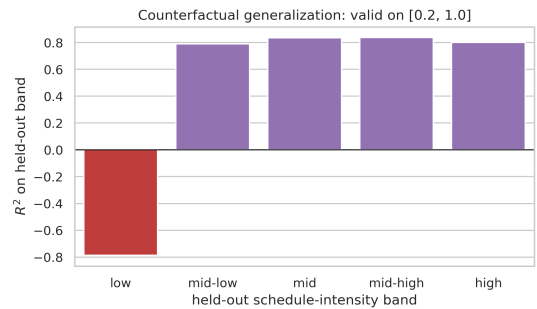


Fig. 6. Counterfactual generalization across held-out schedule-intensity bands

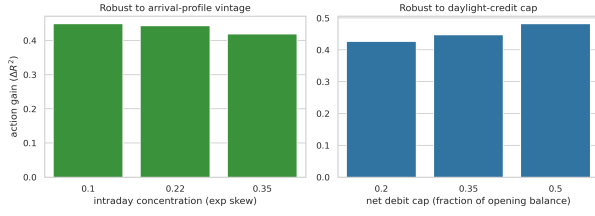


Fig. 7. Sensitivity of the action gain to environment calibration parameters

V. CONCLUSION

The proposed certification system outputted superior predictive performance relative to the static rules used in current practice, with an F_1 of 0.943 versus 0.861 ($p = 5 \times 10^{-6}$), an R^2 of 0.892 on unsettled-value risk, and conformally calibrated 90% intervals covering 0.895 of realized outcomes. These results demonstrate that the system was highly accurate in identifying the mornings on which a bank's reserves and planned schedule would fail to meet its settlement-risk target, allowing buffer decisions to be made with quantified confidence rather than fixed rules.

The benchmark analysis confirms the mechanism behind this performance: conditioning on the planned release schedule contributes $+0.556 R^2$ ($p = 6 \times 10^{-6}$), while an oracle granted complete visibility of every counterparty's queues and balances provides no additional accuracy ($p = 0.39$). The information that determines a bank's settlement outcome is not hidden in the system; it is contained in the bank's own planned actions, validating the analytic bound of Section III-B and the paper's central reframing of intraday liquidity management as a control problem rather than an estimation problem.

Ultimately, the findings suggest that action-conditional prediction with calibrated uncertainty forms the foundation of a robust and deployable instrument for intraday liquidity management, one whose advantage is largest in the ample-reserve regime toward which Federal Reserve policy is moving, and which therefore addresses the operational gap where static rules degrade most.

VI. LIMITATIONS AND FUTURE WORK

While the proposed system achieves strong performance, certain constraints and assumptions merit discussion.

Despite the environment being calibrated entirely from public FDIC, Fedwire, and Federal Reserve data, all mechanism-level results are simulation-based, as no participant-level intraday payment data is publicly available. Validation is therefore against an aggregate empirical relationship rather than the microdynamics themselves. However, the system is designed so that retraining on a bank's internal Fedwire data is a drop-in substitution; future work in collaboration with an institution or regulator holding such data could field-test the certifier directly.

Additionally, the study evaluates fixed policy families without strategic response, as other banks do not react to the ego bank's schedule, and whether release-on-arrival survives as an

equilibrium under mutual optimization remains the classical payment-game question. Future research could embed the certifier within a multi-agent equilibrium and train its internal representation on realized cost rather than prediction error [11], allowing the deviation between the decision-optimal and statistically optimal estimate to be measured directly.

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